



# Explosion-Proof Quadruped Robot – X5



■ Product Overview

The robot features comprehensive advantages including explosion-proof certification, lightweight design, high integration, and exceptional operational stability. It is specifically engineered to meet the safety requirements of mission-critical industrial environments. The body is constructed using aerospace-grade alloy materials, delivering high strength and outstanding reliability. Equipped with an upgraded wheel-leg hybrid locomotion structure and a newly developed motion control algorithm, the X5 combines superior mobility with excellent terrain adaptability. It is capable of operating across a wide range of complex industrial terrains and specialized operational scenarios, making it widely applicable in emergency response and industrial inspection sectors.

■ Application Scenarios

The X5 is widely used in petroleum, chemical, natural gas, steel, and liquor industries. It supports unmanned inspection and reconnaissance operations and is suitable for deserts, forests, mountains and Gobi environments.

■ Basic Parameters

|                                 |                                                        |
|---------------------------------|--------------------------------------------------------|
| Dimensions                      | ≤1200mm×750mm×950mm (L×W×H, subject to actual product) |
| Weight                          | ≤125kg (subject to actual product)                     |
| Explosion-Proof Rating          | Ex IIB T4 Gb                                           |
| Protection Rating               | IP66                                                   |
| Degrees of Freedom              | 16+2                                                   |
| Drive Type                      | Wheel-leg hybrid (articulated system)                  |
| Straight-line Speed             | ≤2.0m/s                                                |
| Braking Distance                | ≤0.5m                                                  |
| Minimum Passage Width           | Flat: 1000mm                                           |
| Payload                         | 30kg                                                   |
| Climbing Ability                | ≤30°                                                   |
| Stair Climbing                  | Step height ≤20 cm, width ≥25cm                        |
| Obstacle Negotiation Capability | ≤40cm                                                  |
| Operating Temperature           | -20°C~+60°C                                            |
| Wading Depth                    | ≤15cm                                                  |
| Charging Time                   | ≤3h                                                    |
| Operating Endurance             | ≥2 hours                                               |
| Main Navigation Mode            | LiDAR SLAM + IMU + Visual Positioning                  |
| Positioning Accuracy            | ≤±5cm                                                  |

■ Standard Functions

- Autonomous Obstacle Avoidance
  - Emergency stop
  - Autonomous Charging
  - Auxiliary Lighting
  - Status Indication
- Audio-visual Alarm
  - Edge Computing
  - Autonomous Inspection
  - Voice Intercom
  - Voice Broadcast
- Real-time Monitoring
  - Voice Interaction
  - Video Surveillance
  - Self-diagnosis
- One-key Return
  - Offline Inspection
  - Monitoring Platform
  - Fault Diagnosis

■ Optional Functions

- Flexible optional add-ons

■ Scenario Algorithms

- Personnel Detection Scenario Algorithms
  - Anomaly Detection Scenario Algorithms
- Power-related Scenario Algorithms
  - Petrochemical Scenario Algorithms
- Public Safety Scenario Algorithms
  - ...



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